CHAPTER-5

What are the characteristics of nichrome that make it suitable for a heating element?

High oxidation resistance/Ability to provide heat energy for a long time in a red hot state.

$H=I^2Rt$--Joule's law.

If voltage is constant, when the resistance in the circuit is decreased, the current increases. Hence the quantity of heat produced increases.

$$P = I^2 R = \frac{V^2}{R} = VI$$

$$H = Pt$$

How much heat energy is produced when an electric heater with 600 W power operates for 20 minutes?

t=20x60=1200 E=Pxt 600x1200=720000j

One kilowatt hour or one unit of electric energy is the energy consumed by an appliance with 1000 W.

What can we do to reduce electricity consumption at home?

Use energy efficient electrical appliances./Turn off switches immediately after use.

What can be done to reduce carbon footprint?

• Reduce domestic energy consumption./Avoid wasting food.

CHAPTER-6

The phenomenon of inducing an emf across a conductor due to a change in the magnetic flux linked with the conductor is electromagnetic induction.

AC generator : Slip rings, Brushes

DC generator : ..split ring,brushes

Thermal power station	Heat energy ⟶ Electric energy
Nuclear power station	..... ⟶ .....
Hydroelectric power station	..... ⟶ .....

Nuclear energy---> Heat energy----> Electric energy

Potential energy -->Kinetic energy----> Electric energy

What are the three factors that influence the quantity of heat produced when current flows through a conductor?

The quantity of heat produced when current flows through a conductor is influenced by the square of the current the resistance of the conductor and the time for which the current flows.

Stepup transformer	Stepdown transformer
Thick wires are used in the primary.	• Thin wires are used in the primary
• The primary has lesser number of turns than the secondary.	**The primary has a greater numbe of turns than the secondary.**

At what voltage is electricity generated in a power station? 11kv

Where are stepdown transformers used? substation

What is the voltage of electricity supplied to houses?220v

How many wires reach the watt hour meter?2

Which device is connected between the watt hour meter and the main switch?fuse

Red coloured wires are used commonly for the phase line, black for the neutral line and green for the earth line.

Fuse wire has a relatively low melting point.

there is excessive current in a circuit due to short circuit or overload,the MCB automatically operates and disconnects the circuit ie., it trips.

In household electric circuits, branch circuits start after the ELCB.

Do not handle electrical appliances or operate switches with wet hands. Plug into or unplug from a socket only after turning off the switch.

First Aid for Electric Shock

Rub the body to increase blood circulation and raise body temperature.

Administer artificial respiration.

CHAPTER-7

Simple machines are devices that change the magnitude of the effect of force or the direction of the force or both.

Mechanical advantage is only a ratio. It has no unit.

A lever is a rigid rod that can rotate around a fixed point called fulcrum

The force we apply to a simple machine is the effort (E). The force the simple machine has to overcome is the load (L).

If the fulcrum is in between the load and the effort, it is a first order lever.

Eg:common balance, scissors, seesaw.

If the load comes in between the effort and the fulcrum, it is a second order lever.

Eg:wheelbarrow,nutcracker

If the effort comes in between the load and the fulcrum, it is a third order lever.

Eg:sugartong,knife

mechanical advantage---first oder=more than 1,equal to 1 or less than 1.

mechanical advantage---second oder=more than 1.

mechanical advantage---third oder=less than 1.

What is the advantage of using friction less fixed pulley to lift objects? Choose the correct option from those given below.

(can reduce the magnitude of the applied force / can change the direction of the applied force)

There is no gain in work by using simple machines

Is there a mechanical advantage while using an inclined plane?yes.

What is the work done to lift a 600 N load by 3 m? w=fxd=600x3=1800j